Retrieval over the eCFR: Seven Rounds of Measured Experiments

Technical whitepaper — ecfr-mcp retrieval bake-off, 2026-10-12/13

Abstract

We built a retrieval layer over a Markdown corpus of the U.S. Code of Federal Regulations (2,278 current-version benefit-program sections) and measured nine configurations against a single frozen 50-query golden set. The simplest production baseline (deterministic citation lookup + dense vectors) beat BM25, hybrid fusion, chunking, and a cross-encoder reranker. Two LLM-assisted techniques then moved the needle: HyDE query expansion (+8pp recall@1) and listwise LLM reranking (+12pp recall@1 over baseline, natural-query recall@3/5 perfect). Total experiment spend: about $0.18. Every number below is measured on the same frozen set — no query was ever edited to make a system pass.

1. Corpus and method

• **Corpus:** `ctrimm/ecfr-markdown` — the eCFR as Markdown, one file per section, current as of 2026-10-08. Indexed subset: 2,278 current-version sections across 12 benefit programs (SNAP, WIC, TANF, Medicaid/CHIP, LIHEAP, SSI, housing vouchers, Lifeline, child care, UI, school meals, Summer EBT).

• **Embeddings:** `BAAI/bge-base-en-v1.5` (768-dim), brute-force cosine.

No instruction prefixes — round 1 showed bge v1.5 was not trained with them and they actively hurt (exact-citation queries ranked ~1270th of 2,278, worse than random).

• **Golden set:** 50 queries frozen *before any retrieval ran* — 5 exact citations ("7 CFR 273.9"), 33 natural questions, 12 paraphrases, stratified across 14 program areas. Sealed: queries are added, never edited to fit the system.

• **Metric:** recall@1 / @3 / @5, overall and per query type.

2. The production baseline

A query matching `\d+ CFR [\w.]+` is a **deterministic lookup**, not retrieval — a query that IS a citation resolves by key (recall 1.00 by construction, and correct by design). Everything else ranks by cosine similarity over whole-section vectors. BM25 (SQLite FTS5, title 3x weight) was built and kept as a fallback/debug leg, but RRF fusion never beat vectors+citation on any k, so BM25 is not in the ranking path.

3. Seven rounds (chart: progression.png, recall_all.png)

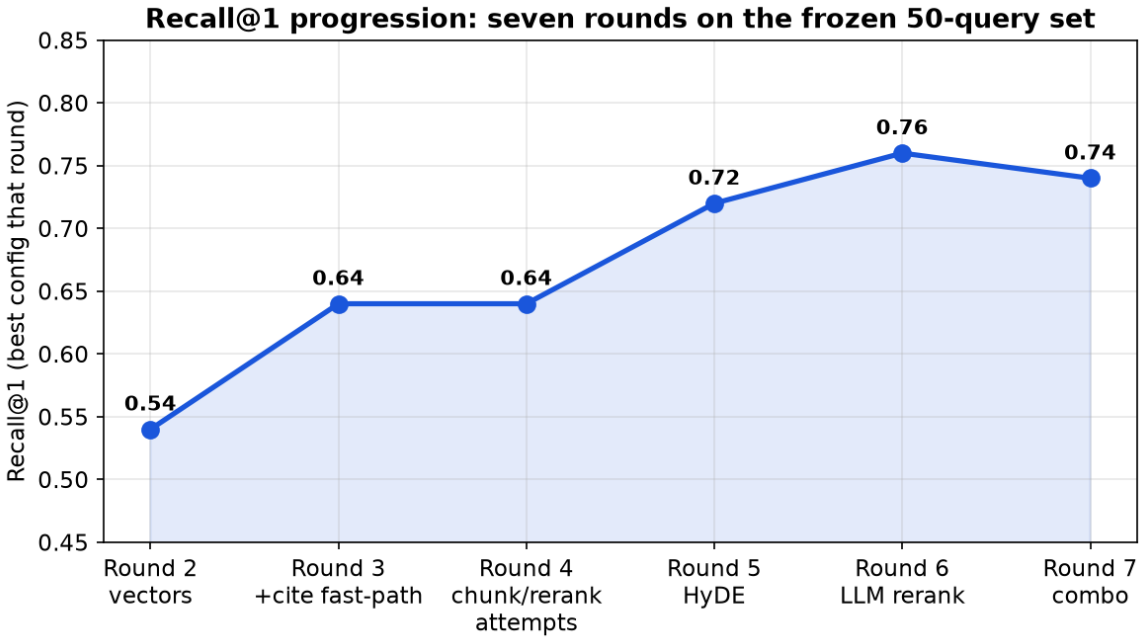


Figure 1 — Recall@1 progression across the seven rounds.

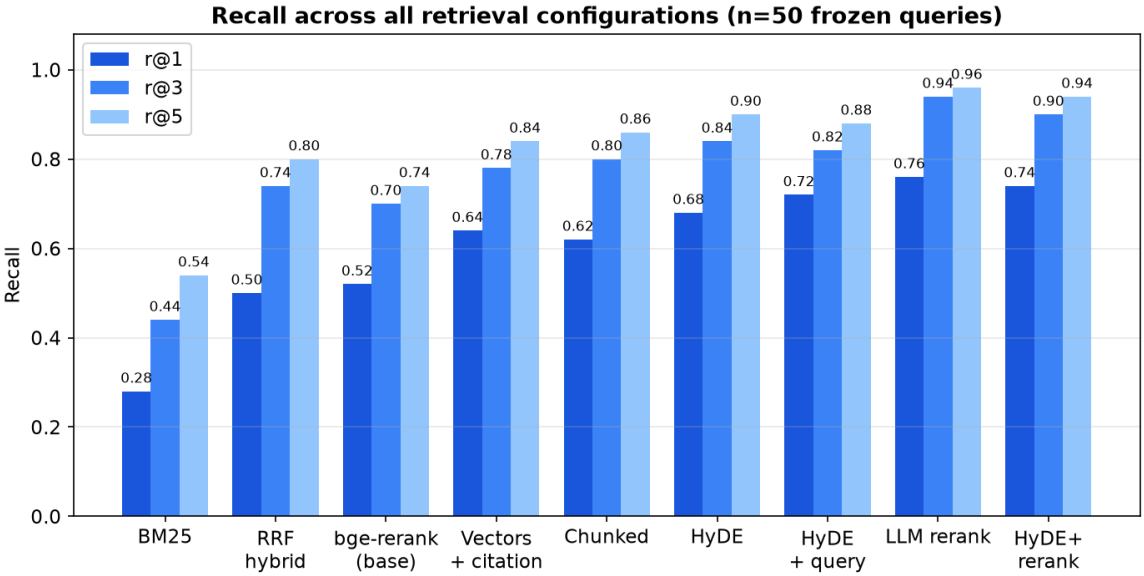


Figure 2 — Recall across all nine configurations.

Round	System	r@1	r@3	r@5	Cost
2	Vectors, no prefixes	0.540	0.720	0.800	$0
3a	BM25 alone	0.280	0.440	0.540	$0
3b	RRF hybrid (best k)	0.500	0.760	0.840	$0
3c	**Vectors + citation fast-path**	**0.640**	**0.780**	**0.840**	$0

4a	Chunked index (6,222 chunks, best-chunk)	0.620	0.800	0.860	$0
4b	bge-reranker-base, sections	0.520	0.700	0.740	$0
4c	bge-reranker-base, chunks	0.520	0.700	0.760	$0
5a	HyDE passage only	0.680	0.840	0.900	$0.006
5b	**HyDE + query averaged**	**0.720**	**0.820**	**0.880**	$0.006
6	**LLM rerank, query candidates**	**0.760**	**0.940**	**0.960**	$0.09
7	HyDE candidates + LLM rerank	0.740	0.900	0.940	$0.18

Per-type recall@1 for the three leaders (chart: recall_by_type.png):

System	natural (n=33)	paraphrase (n=12)	exact (n=5)
Vectors + citation	0.667	0.417	1.000
HyDE + query	0.758	0.500	1.000
LLM rerank	0.788	0.583	1.000

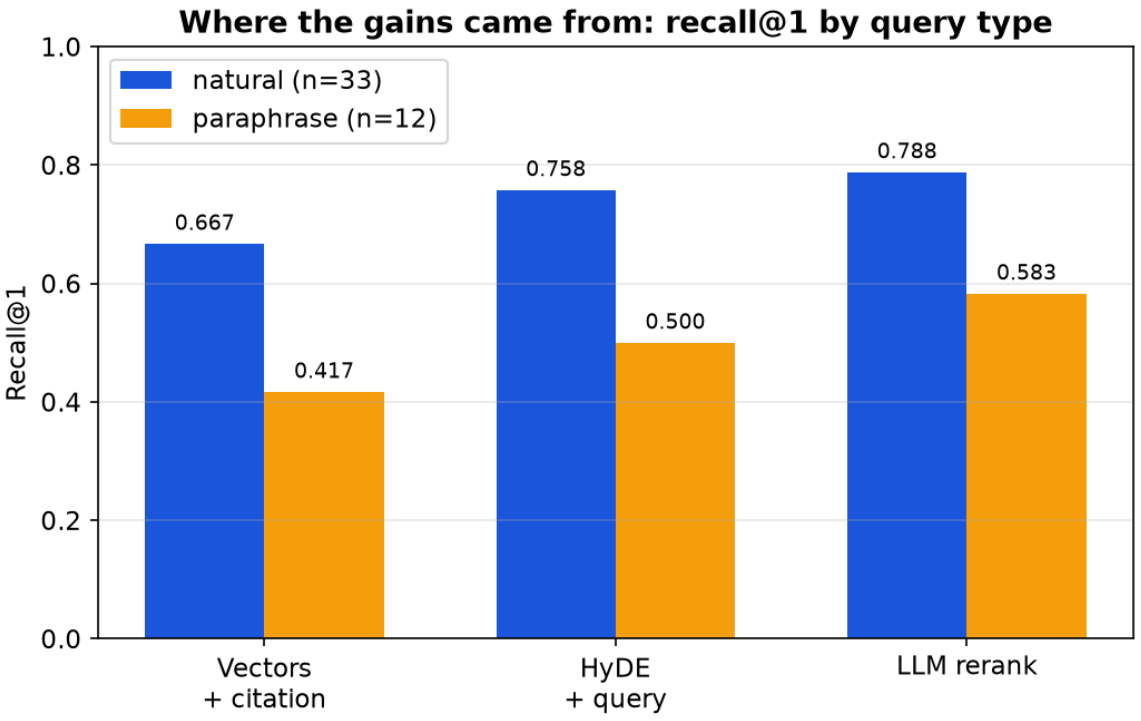


Figure 3 — Recall@1 by query type for the three leaders.

LLM rerank additionally reaches natural r@3/r@5 = **1.000** (perfect).

4. Findings

1. **Prefixes hurt.** bge-base-en-v1.5 was not trained with `query:` / `passage:` prefixes; both sides landed out-of-distribution. Removing them: 0.40 -> 0.54 r@1.

2. **A citation is a lookup.** Short citation strings carry almost no semantic signal; dense retrieval ranked "20 CFR 416.1100" 592nd. The deterministic fast-path fixes the whole category.

3. **BM25-first loses.** RRF fusion (k=5..60) never beats vectors + citation on any cutoff. BM25 stays as fallback only.

4. **39.7% of sections exceed bge's 512-token limit** and were silently truncated — real data loss, but chunking (450-token windows, 64 overlap, title-prepended, section = best chunk) measured as a wash (0.62/0.80/0.86): it fixed truncation in principle while diluting topical signal on paraphrase queries (r@1 0.417 -> 0.250).

5. **bge-reranker-base hurts in both configurations** (0.52/0.70/0.74 on sections, 0.52/0.70/0.76 on chunks). Reranking chunks eliminated the truncation confound, so the cause is model/domain mismatch — MS MARCO training rewards surface lexical overlap and promotes distractor docs (e.g. the OMB control-number list that cites everything).

6. **HyDE is the first real win** (+8pp r@1, $0.0056 total): a flash model writes a 4-6 sentence CFR-style passage answering the query; embedding the passage (alone, or averaged with the query vector) beats the query vector alone. HyDE-only is strongest on paraphrase (0.750); the average is the balanced best.

7. **Listwise LLM reranking wins overall** (gemini-2.5-flash-lite scores top-50 chunks 10-at-a-time, section = best chunk): 0.76/0.94/0.96 for $0.09. Operational note: deepseek-v4-flash via SiliconFlow returned intermittent null-content HTTP 200s in streaks and was abandoned after gemini went 5/5 clean on identical prompts.

8. **The combo does not stack.** HyDE candidates + LLM rerank scored 0.74/0.90/0.94 — worse everywhere than query candidates + rerank. The reranker does the heavy lifting; once it sees the top-50, the candidate source barely matters. Rejected (extra LLM call, no gain).

5. Cost analysis (chart: cost_frontier.png)

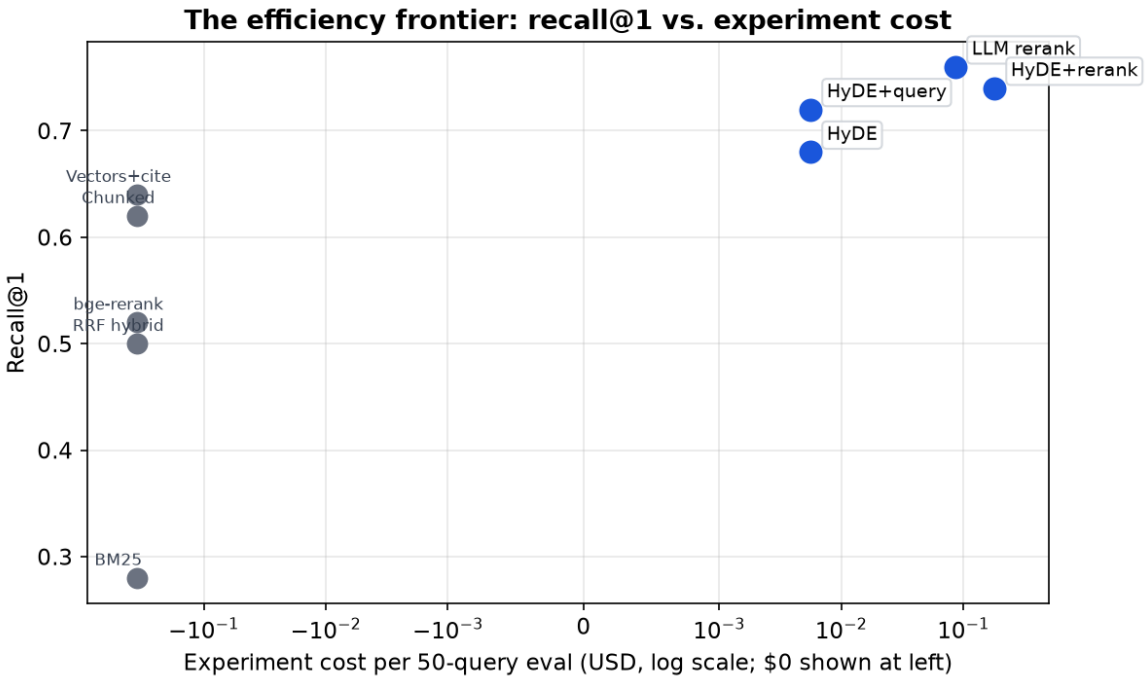


Figure 4 — The efficiency frontier: recall@1 vs experiment cost.

The efficiency frontier is steep and short: $0.18 of total experiment spend bought +12pp r@1 / +16pp r@3 over the free baseline. Per-query serving costs at the winning config: ~6 flash-model calls, roughly

$0.002 and 30-60s sequential latency (batchable). Whether that belongs in the default serving path is a product decision — see deployment tiers below.

6. Deployment: env-var-gated tiers (proposed MCP server design)

The server starts in the tier its environment supports — no flags, no config files:

• **Free tier** (no key): citation fast-path + local vectors. 0.64/0.78/0.84, fully offline after install, $0/query.

• **Best tier** (`OPENROUTER_API_KEY` set in the user's shell): adds the listwise LLM rerank over top-50 chunks. 0.76/0.94/0.96, ~$0.002/query.

• **Override:** `ECFR_MCP_TIER=free|best` pins a tier regardless of the key (e.g. force free for latency-sensitive use, or test best without exporting a key globally).

Degradation is graceful: if the key is absent or the LLM call fails, the server serves the free tier rather than erroring. The golden set and all seven experiment scripts ship in the repo so any tier choice stays re-measurable.

7. Limitations

n=50 is small — differences under ~0.04 are noise, and several "wins" should be re-confirmed on a larger set before hardening into lore. All embedding work ran CPU-only on a 7GB box (bge-reranker-v2-m3 OOM-killed there; the LLM judge replaced it). One corpus snapshot (eCFR 2026-10-08);

version-chaining retrieval is built but not evaluated here.

8. Reproducibility

Repo: `ctrimm/ecfr-mcp` (private). `golden/eval.json` (frozen set), `golden/eval.py`, `golden/eval_chunks.py`, `golden/exp_rerank.py`, `golden/exp_rerank_chunks.py`, `golden/exp_hyde.py`, `golden/exp_llm_rerank.py`; `golden/results-2026-10-12.md` (all rounds); `DECISION_LOG.md` (each data-backed decision with evidence and date).